THE EXAM WORKBOOK

Grade 10 General

افهم الفكرة. أتقن الحل.

Exponents. Polynomials. Quadratics.

25 learning outcomes

101 focused questions

60 minute online mock

Name / الاسم _____ Class / الصف _____

A short route to confident practice

READ → REASON → CALCULATE → CHECK. Calculator guides on the website use fx-991EX ClassWiz.

اقرأ القاعدة، حلّ بنفسك، ثم راجع خطواتك. لكل مخرج 4 أسئلة مختارة، ولخواص ضرب القوى 5 أسئلة لتغطية الأفكار المتعددة.

Your topic map

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Selected from the uploaded Reveal textbook, printed pp. 569–738, using the supplied 2026–2027 Grade 10 General coverage. Written tasks and original diagrams come from the book. MCQs include exam-style adaptations and labelled concept/example practice. This is a study selection, not an official exam or a prediction. The official coverage allows a scientific calculator and specifies 150 minutes / 100 marks. Online mock: 20 MCQs / 60 minutes. Calculator: fx-991EX ClassWiz.

معطيات المقالية من الكتاب، والاختيار من متعدد يتضمن تطبيقات موثقة للمفاهيم والأمثلة. لا يمكن ضمان ورود أسئلة بعينها في الاختبار النهائي.

01 / CHOOSE & CHECK

Q1 — Q2

Q1 FOIL binomial products

ضرب ثنائيي الحد

Remember Multiply all four pairs, then combine the two middle terms.

1.1 MCQ

Which option gives the product?

$$(c+3)\,(3c-5)$$

Reveal · p. 649 · 1 · MCQ adaptation

A $3c^2+4c$

B $3c^2+4c+15$

C $(3c^2+4c)-15$

D $4c-15$

1.2 MCQ

Which option gives the product?

$$(g+10)\,(2g-5)$$

Reveal · p. 649 · 2 · MCQ adaptation

A $2g^2+15g$

B $(2g^2+15g)-50$

C $2g^2+15g+50$

D $15g-50$

1.3 MCQ

Which option gives the product?

$$(5a+3)\,(6a+5)$$

Reveal · p. 649 · 3 · MCQ adaptation

A $(30a^2+43a)+15$

B $30a^2+43a$

C $30a^2+43a-15$

D $43a+15$

1.4 MCQ

Which option gives the product?

$$(4x+1)\,(6x+3)$$

Reveal · p. 649 · 4 · MCQ adaptation

A $24x^2+18x$

B $24x^2+18x-3$

C $24x^2+3$

D $(24x^2+18x)+3$

Q2 Squares of sums and differences

مربع مجموع وفرق

Remember (a ± b)² = a² ± 2ab + b². Do not omit the middle term.

2.1 MCQ

Which option gives the product?

$$(a+10)(a+10)$$

Reveal · p. 657 · 1 · MCQ adaptation

A a^2+20a

B $(a^2+20a)+100$

C $a^2+20a-100$

D $20a+100$

2.2 MCQ

Which option gives the product?

$$(b-6)(b-6)$$

Reveal · p. 657 · 2 · MCQ adaptation

A $(b^2-12b)+36$

B b^2-12b

C $b^2-12b-36$

D $36-12b$

2.7 MCQ

Which identity correctly expands the square of a sum?

$$(a+b)^2$$

Reveal · p. 657 · Application of 1 · Concept application

A $a^2+2ab+b^2$

B a^2+b^2

C $a^2-2ab+b^2$

D $2a+2b$

2.8 MCQ

Which missing term makes this identity true?

$$(x-5)^2=x^2+\square+25$$

Reveal · p. 657 · Application of 1 · Concept application

A $-5x$

B -25

C $-10x$

D $10x$

01 / CHOOSE & CHECK

Q3 — Q4

Q3 Sum and difference products

حاصل ضرب مجموع وفرق

Remember (a+b)(a−b)=a²−b²; the middle terms cancel.

3.1 MCQ

Which option gives the product?

$$(u+3)(u-3)$$

Reveal · p. 658 · 15 · MCQ adaptation

A u^2-9 B u^2
C u^2+9 D -9

3.2 MCQ

Which option gives the product?

$$(b+7)(b-7)$$

Reveal · p. 658 · 16 · MCQ adaptation

A b^2 B b^2+49
C -49 D b^2-49

3.3 MCQ

Which option gives the product?

$$(2+x)(2-x)$$

Reveal · p. 658 · 17 · MCQ adaptation

A $-x^2$ B $-x^2-4$
C $4-x^2$ D 4

3.4 MCQ

Which option gives the product?

$$(4+x)(4-x)$$

Reveal · p. 658 · 18 · MCQ adaptation

A $-x^2$ B $16-x^2$
C $-x^2-16$ D 16

Q4 Factor using distribution

التحليل بخاصية التوزيع

Remember Take the greatest common factor from every term; check by multiplication.

4.1 MCQ

Which option is the complete factorisation of the expression?

$$16t-40y$$

Reveal · p. 665 · 1 · MCQ adaptation

A $-16t-40y$ B $16t+40y$
C $-2t+5y$ D $-8\,(-2t+5y)$

4.2 MCQ

Which option is the complete factorisation of the expression?

$$30v+50x$$

Reveal · p. 665 · 2 · MCQ adaptation

A $-30v+50x$ B $30v-50x$
C $10\,(3v+5x)$ D $3v+5x$

4.3 MCQ

Which option is the complete factorisation of the expression?

$$2k^2+4k$$

Reveal · p. 665 · 3 · MCQ adaptation

A $2k\,(2-k)$ B $2k\,(k+2)$
C $2k\,(k-2)$ D $k\,(k+2)$

4.4 MCQ

Which option is the complete factorisation of the expression?

$$5z^2+10z$$

Reveal · p. 665 · 4 · MCQ adaptation

A $5z\,(z+2)$ B $5z\,(2-z)$
C $5z\,(z-2)$ D $z\,(z+2)$

01 / CHOOSE & CHECK

Q5 — Q6

Q5 Analyse monic trinomial factors

تحليل ثلاثيات الحدود ومعاملاتها

Remember For (x+r)(x+s), r+s=b and rs=c. Enumerate integer factor pairs.

5.2 MCQ

Jamaall: (x+2)(x−8). Charles: (x−2)(x+8). Who correctly factored x²+6x−16?

Reveal · p. 674 · 74 · MCQ adaptation

A Neither student

B Charles only

C Jamaall only

D Both students

5.3 MCQ

Which option gives all values of k for which the polynomial factors using integers?

$$x^2 + kx - 19$$

Reveal · p. 674 · 75 · MCQ adaptation

A $k \in \{-18, 18\}$

B $k \in \{-20, 20\}$

C $k \in \{-19, 19\}$

D $k = 18$

5.5 MCQ

Which option gives all values of k for which the polynomial factors using integers?

$$x^2 - 8x + k, \quad k > 0$$

Reveal · p. 674 · 77 · MCQ adaptation

A $k \in \{8, 16\}$

B $k \in \{1, 2, 3, 4\}$

C $k \in \{7, 12, 15, 16\}$

D $k \in \{7, 12, 15\}$

5.6 MCQ

Which option gives all values of k for which the polynomial factors using integers?

$$x^2 - 5x + k, \quad k > 0$$

Reveal · p. 674 · 78 · MCQ adaptation

A $k = 6$

B $k \in \{4, 6\}$

C $k \in \{4, 5, 6\}$

D $k \in \{1, 5\}$

Q6 Factor non-monic trinomials

تحليل ثلاثيات الحدود غير الأحادية

Remember Find two numbers with product ac and sum b; split the middle term and group.

6.1 MCQ

Which option is the complete factorisation of the expression?

$$5x^2 + 34x + 24$$

Reveal · p. 672 · 21 · MCQ adaptation

A $(6 - x)(5x + 4)$

B $(x + 6)(5x + 4)$

C $(x - 6)(5x + 4)$

D $(4 - 5x)(x + 6)$

6.2 MCQ

Which option is the complete factorisation of the expression?

$$2x^2 + 19x + 24$$

Reveal · p. 672 · 22 · MCQ adaptation

A $(x + 8)(2x + 3)$

B $(8 - x)(2x + 3)$

C $(x - 8)(2x + 3)$

D $(3 - 2x)(x + 8)$

6.3 MCQ

Which option is the complete factorisation of the expression?

$$4x^2 + 22x + 10$$

Reveal · p. 672 · 23 · MCQ adaptation

A $(2x + 2)(2x + 5)$

B $-2(x + 5)(2x + 1)$

C $4(x + 5)(2x + 1)$

D $2(x + 5)(2x + 1)$

6.4 MCQ

Which option is the complete factorisation of the expression?

$$4x^2 + 38x + 70$$

Reveal · p. 672 · 24 · MCQ adaptation

A $-2(x + 7)(2x + 5)$

B $4(x + 7)(2x + 5)$

C $2(x + 7)(2x + 5)$

D $(2x + 7)(2x + 10)$

01 / CHOOSE & CHECK

Q7

Q7 Difference of squares

تحليل الفرق بين مربعين

Remember $A^2-B^2=(A-B)(A+B)$. Factor repeatedly when possible.

7.1 MCQ

Which option is the complete factorisation of the expression?

$$q^2 - 121$$

Reveal · p. 679 · 1 · MCQ adaptation

A $(q - 11)(q + 11)$
B $(-q - 11)(q + 11)$
C $(q + 11)(q + 11)$
D $(11 - q)(q - 11)$

7.2 MCQ

Which option is the complete factorisation of the expression?

$$-k^4 + r^4$$

Reveal · p. 679 · 2 · MCQ adaptation

A $(k + r)^2 (k^2 + r^2)$
B $(-k - r)(k + r)(k^2 + r^2)$
C $-(k + r)(k^2 + r^2)(-k + r)$
D $(k + r)(k^2 + r^2)(-k + r)$

7.3 MCQ

Which option is the complete factorisation of the expression?

$$w^4 - 625$$

Reveal · p. 679 · 3 · MCQ adaptation

A $(-w - 5)(w + 5)(w^2 + 25)$
B $(w + 5)(w + 5)(w^2 + 25)$
C $(w + 5)(w^2 + 25)(w - 5)$
D $-(w + 5)(w^2 + 25)(w - 5)$

7.4 MCQ

Which option is the complete factorisation of the expression?

$$r^2 - 9t^2$$

Reveal · p. 679 · 4 · MCQ adaptation

A $(-r - 3t)(r + 3t)$
B $(r + 3t)(r - 3t)$
C $(r + 3t)(r - 3t + 6t)$
D $(-r + 3t)(r - 3t)$

01 / CHOOSE & CHECK

Q8

Q8

Divide monomials

قسمة وحيدات الحد

Remember Subtract exponents when dividing; multiply every exponent when raising a quotient to a power.

8.2 MCQ

Simplify. Assume no denominator equals zero.

$$\left(\frac{p^2t^7}{10}\right)^3$$

Reveal · p. 583 · 5 · MCQ adaptation

A $\frac{p^5t^{10}}{1000}$

B $\frac{p^6t^{21}}{10}$

C $\frac{p^6t^{21}}{1000}$

D $\frac{p^6t^{21}}{30}$

8.3 MCQ

Simplify. Assume no denominator equals zero.

$$\frac{a^7b^8c^8}{a^5bc^7}$$

Reveal · p. 583 · 6 · MCQ adaptation

A $a^{12}b^9c^{15}$

B a^2b^7c

C a^2b^8c

D $a^2b^7c^{15}$

8.7 MCQ

Which expression is equivalent when b≠0?

$$\left(\frac{a}{b}\right)^3$$

Reveal · p. 583 · Application of 3 · Concept application

A $\frac{3a}{3b}$

B $\frac{a^3}{b^3}$

C $\frac{a^3}{b}$

D $\frac{a}{b^3}$

8.8 MCQ

Which simplified expression is correct for x≠0?

$$\frac{x^7}{x^3}$$

Reveal · p. 583 · Application of 3 · Concept application

A x^{10}

B x^{21}

C $x^{7/3}$

D x^4

01 / CHOOSE & CHECK

Q9 — Q10

Q9 Negative and zero exponents

الأسس السالبة والصفرية

Remember For a≠0, $a^0=1$ and $a^{-n}=1/a^n$. Move only factors with negative exponents.

9.1 MCQ

Which option is equivalent to the expression in simplified form?

$$\frac{3m^{-3}t^4p^2}{12t^4}$$

Reveal · p. 591 · 11 · MCQ adaptation

A $-\frac{m^3p^2}{4}$

B $\frac{m^3p^2}{4}$

C $\frac{p^2}{4m^3}$

D $\frac{p^2}{4m^3t^8}$

9.2 MCQ

Which option is equivalent to the expression in simplified form?

$$\left(\frac{3t^6u^2v^5}{9tuv^{21}}\right)^0$$

Reveal · p. 591 · 12 · MCQ adaptation

A 0

B 1

C -1

D $\frac{t^5u}{3v^{16}}$

9.7 MCQ

Which statement holds for every nonzero real number a?

$$a^0$$

Reveal · p. 591 · Application of 11 · Concept application

A 1

B 0

C a

D -1

9.8 MCQ

Which expression uses positive exponents only? Assume x≠0.

$$x^{-4}$$

Reveal · p. 591 · Application of 11 · Concept application

A x^4

B $-\frac{1}{x^4}$

C $\frac{1}{x^4}$

D $-x^4$

Q10 Rational exponents

الأسس النسبية والجذور

Remember a^(1/n) is the principal nth root of a. Evaluate the root before any other power.

10.1 MCQ

Evaluate.

$$\left(\frac{1}{16}\right)^{1/4}$$

Reveal · p. 599 · 9 · MCQ adaptation

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $\frac{1}{4}$

D 2

10.2 MCQ

Evaluate.

$$\sqrt[5]{3125}$$

Reveal · p. 599 · 10 · MCQ adaptation

A 5

B 625

C 25

D -5

10.5 MCQ

Evaluate.

$$\sqrt[6]{4096}$$

Reveal · p. 599 · 13 · MCQ adaptation

A 64

B 4

C 8

D -4

10.6 MCQ

Evaluate.

$$\left(\frac{16}{625}\right)^{1/4}$$

Reveal · p. 599 · 15 · MCQ adaptation

A $\frac{2}{5}$

B $\frac{4}{5}$

C $\frac{4}{25}$

D $-\frac{2}{5}$

01 / CHOOSE & CHECK

Q11

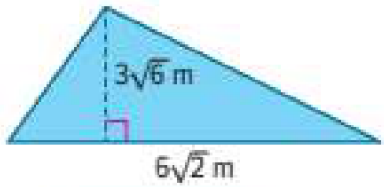
Q11 Products of square roots
ضرب الجذور التربيعية

Remember √a·√b=√(ab) for a,b≥0. Extract perfect-square factors; area uses the full formula.

11.1 MCQ

The base of a triangle is 6√2 m and its height is 3√6 m. What is its area?

Reveal · p. 609 · 67 · MCQ adaptation

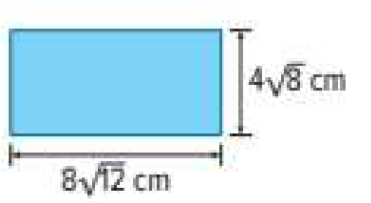


A $18\sqrt{3}$ B $36\sqrt{3}$
C $18\sqrt{2}$ D $9\sqrt{3}$

11.2 MCQ

A rectangle is 8√12 cm long and 4√8 cm wide. What is its area?

Reveal · p. 609 · 68 · MCQ adaptation



A $64\sqrt{6}$ B $32\sqrt{6}$
C $128\sqrt{3}$ D $128\sqrt{6}$

11.3 MCQ

Use √(h·k) to find the geometric mean of 32 and 14 in simplified radical form.

Reveal · p. 609 · 69 · MCQ adaptation

A $16\sqrt{7}$ B $8\sqrt{14}$
C $8\sqrt{7}$ D $4\sqrt{7}$

11.4 MCQ

Fernanda has paint to cover 5 square yards of a cubic prop. Use s=√(A/6). What is the longest side length, to the nearest tenth of a yard?

Reveal · p. 609 · 70 · MCQ adaptation

A $\frac{1}{2}$ B $\frac{9}{10}$
C $\frac{4}{5}$ D $\frac{6}{5}$

01 / CHOOSE & CHECK

Q12

Q12 Quotients of radicals
قسمة الجذور وتبسيطها

Remember √(a/b)=√a/√b for a≥0,b>0. Simplify and rationalise the denominator. For cube roots, extract cubes.

12.1 MCQ

Simplify.

$$\sqrt{\frac{192}{216}}$$

Reveal · p. 607 · 15 · MCQ adaptation

A $\frac{8}{9}$
B $\frac{2\sqrt{2}}{9}$
C $\frac{\sqrt{2}}{3}$
D $\frac{2\sqrt{2}}{3}$

12.2 MCQ

Simplify.

$$\sqrt{\frac{88}{18}}$$

Reveal · p. 607 · 16 · MCQ adaptation

A $\frac{44}{9}$
B $\frac{2\sqrt{11}}{9}$
C $\frac{2\sqrt{11}}{3}$
D $\frac{\sqrt{11}}{3}$

12.3 MCQ

Simplify.

$$\sqrt{\frac{84}{121}}$$

Reveal · p. 607 · 17 · MCQ adaptation

A $\frac{84}{121}$
B $\frac{2\sqrt{21}}{11}$
C $\frac{2\sqrt{21}}{121}$
D $\frac{\sqrt{21}}{11}$

12.4 MCQ

Simplify.

$$\sqrt{\frac{75}{20}}$$

Reveal · p. 607 · 18 · MCQ adaptation

A $\frac{\sqrt{15}}{2}$
B $\frac{15}{4}$
C $\frac{\sqrt{15}}{4}$
D $\frac{\sqrt{55}}{2}$

01 / CHOOSE & CHECK

Q13 — Q14

Q13 Add and subtract radicals
جمع الجذور وطرحها

Remember Simplify the radicals first; combine coefficients only when their radicands match.

13.1 MCQ

Simplify.

$$7\sqrt{7} - 2\sqrt{7}$$

Reveal · p. 615 · 1 · MCQ adaptation

A $9\sqrt{7}$ B $5\sqrt{14}$
C $5\sqrt{7}$ D $14\sqrt{7}$

13.2 MCQ

Simplify.

$$3\sqrt{8} + 7\sqrt{8}$$

Reveal · p. 615 · 2 · MCQ adaptation

A $10\sqrt{2}$ B $20\sqrt{2}$
C 20 D $42\sqrt{2}$

13.5 MCQ

Simplify.

$$3\sqrt{5} - 2\sqrt{20}$$

Reveal · p. 615 · 7 · MCQ adaptation

A $\sqrt{5}$ B $7\sqrt{5}$
C $-\sqrt{5}$ D $\sqrt{15}$

13.6 MCQ

Simplify.

$$3\sqrt{50} - 3\sqrt{32}$$

Reveal · p. 615 · 8 · MCQ adaptation

A $27\sqrt{2}$ B $3\sqrt{2}$
C $9\sqrt{2}$ D $3\sqrt{82}$

Q14 Factoring in context
تطبيق التحليل في مسائل تربيعية

Remember Set a physical model equal to the required height/volume, factor, then reject roots outside its domain.

14.1 MCQ

A ballast bag is dropped from 576 ft. Its height is f(t)=−16t²+576. After how many seconds does it hit the ground?

Reveal · p. 722 · 29 · MCQ adaptation

A 36 B 6
C -6 D 18

14.2 MCQ

Which option is the complete factorisation of the expression?

$$V = 3x^2 - 36x + 108$$

Reveal · p. 722 · 30(a) · MCQ adaptation

A $3(x-6)^2$ B $3(x+6)^2$
C $(3x-6)^2$ D $3(x-12)^2$

14.3 MCQ

Solve by factoring. Check the solutions.

$$x^2 + 9x + 18 = 0$$

Reveal · p. 721 · 15 · MCQ adaptation

A $x = 6, 3$ B $x = -9, -2$
C $x = 0, -9$ D $x = -6, -3$

14.6 MCQ

Solve by factoring. Check the solutions.

$$-x^2 - 12 = -8x$$

Reveal · p. 721 · 18 · MCQ adaptation

A $x = 2, 6$ B $x = -2, -6$
C $x = 3, 4$ D $x = -3, -4$

01 / CHOOSE & CHECK

Q15

Q15 Square root and completing-square methods

الجذر التربيعي وإكمال المربع

Remember Isolate $(x-h)^2$; take both square roots. Complete the square by adding $(b/2)^2$.

15.1 MCQ

Which expression does not belong because it is not a perfect square trinomial?

Reveal · p. 730 · 37 · MCQ adaptation

A $(n^2+n)+\frac{3}{4}$

B $n^2+n-\frac{3}{4}+\frac{3}{4}$

C $n^2+n-\frac{3}{2}+\frac{3}{4}$

D $n^2+\frac{3}{4}$

15.2 MCQ

Create a quadratic equation for which the only solution is 4. Choose a valid equation.

Reveal · p. 730 · 38 · MCQ adaptation

A $x^2-16=0$

B $(x+4)^2=0$

C $x^2-4=0$

D $(x-4)^2=0$

15.7 MCQ

Which operation isolates the square before taking square roots?

$$3(x-2)^2=27$$

Reveal · p. 730 · Application of 37 · Concept application

A Subtract 2 from 27.

B Divide both sides by 3.

C Subtract 3 from both sides.

D Take only the positive root.

15.8 MCQ

Which pair is the complete solution set?

$$(x-2)^2=9$$

Reveal · p. 730 · Application of 37 · Concept application

A $\{-3,3\}$

B $\{-1,5\}$

C $\{5\}$

D $\{-5,1\}$

01 / CHOOSE & CHECK

Q16

Q16 Translations of quadratics
إزاحة الدوال التربيعية

Remember In y=a(x−h)²+k, the vertex is (h,k). x−h shifts right; x+h shifts left.

16.1 MCQ

Describe the translation from the parent function y=x².

$$g(x) = (x-1)^2$$

Reveal · p. 707 · 3 · MCQ adaptation

A Left 1 unit
B Up 1 unit
C Down 1 unit
D Right 1 unit

16.2 MCQ

Describe the translation from the parent function y=x².

$$g(x) = x^2 - 8$$

Reveal · p. 707 · 4 · MCQ adaptation

A Right 8 units
B Left 8 units
C Down 8 units
D Up 8 units

16.7 MCQ

What is the vertex of the translated parabola?

$$y = (x-3)^2 + 2$$

Reveal · p. 707 · Application of 3 · Concept application

A $(3, 2)$
B $(-3, 2)$
C $(3, -2)$
D $(-3, -2)$

16.8 MCQ

Which translation takes y=x² to the given graph?

$$y = (x+4)^2 - 1$$

Reveal · p. 707 · Application of 3 · Concept application

A 4 right and 1 down.
B 4 left and 1 up.
C 1 left and 4 down.
D 4 left and 1 down.

01 / CHOOSE & CHECK

Q17

Q17 Factoring quadratic equations

حل المعادلات التربيعية بالتحليل

Remember Write one side as zero, factor fully, and apply the zero-product property. Check every root.

17.1 MCQ

Solve by factoring. Check the solutions.

$$3b(9b - 27) = 0$$

Reveal · p. 721 · 5 · MCQ adaptation

A $b = 0, -3$
B $b = -3, 3$
C $b = 0, 3$
D $b = 3$

17.2 MCQ

Solve by factoring. Check the solutions.

$$(7x + 3)(2x - 6) = 0$$

Reveal · p. 721 · 6 · MCQ adaptation

A $x = -\frac{7}{3}, 3$
B $x = -\frac{3}{7}, 3$
C $x = \frac{3}{7}, -3$
D $x = -3, 3$

17.3 MCQ

Solve by factoring. Check the solutions.

$$b^2 = -3b$$

Reveal · p. 721 · 7 · MCQ adaptation

A $b = 0, -3$
B $b = -3$
C $b = 0, 3$
D $b = -3, 3$

17.4 MCQ

Solve by factoring. Check the solutions.

$$a^2 = 4a$$

Reveal · p. 721 · 8 · MCQ adaptation

A $a = 4$
B $a = 0, -4$
C $a = -4, 4$
D $a = 0, 4$

01 / CHOOSE & CHECK

Q18

Q18 Dilations and reflections of quadratics

تمدد وانعكاس الدوال التربيعية

Remember A negative outside coefficient reflects across the x-axis; |a|>1 stretches vertically. Simplify an inside coefficient before comparing.

18.1 MCQ

Describe the graph relative to the parent y=x².

$$g(x) = -\frac{1}{3}x^2$$

Reveal · p. 707 · 23 · MCQ adaptation

A Reflect across the x-axis and stretch by 3

B Reflect across the x-axis; vertically compress by 1/3

C Compress vertically by 1/3 without reflection

D Shift down 1/3

18.2 MCQ

Describe the graph relative to the parent y=x².

$$g(x) = (-\frac{2}{3}x)^2$$

Reveal · p. 707 · 24 · MCQ adaptation

A Vertical compression by 4/9; no reflection

B Reflect across the x-axis and compress by 4/9

C Compress vertically by 2/3

D Stretch vertically by 9/4

18.7 MCQ

Which description is correct relative to y=x²?

$$y = -2x^2$$

Reveal · p. 707 · Application of 23 · Concept application

A Translation down 2.

B Reflection in the x-axis and vertical stretch by 2.

C Reflection in the y-axis only.

D Vertical compression by 2.

18.8 MCQ

Which graph is wider than y=x²?

Reveal · p. 707 · Application of 23 · Concept application

A $y = \frac{1}{4}x^2$

B $y = 4x^2$

C $y = -4x^2$

D $y = 2x^2$

01 / CHOOSE & CHECK

Q19 — Q20

Q19 Solve quadratics by graphing

حل المعادلات التربيعية بيانيًا

Remember Real solutions are the x-intercepts of the related graph; a tangent gives a repeated root; no intersection means no real solution.

19.1 MCQ

Which option gives the complete real solution set?

$$x^2 + 7x + 14 = 0$$

Reveal · p. 713 · 1 · MCQ adaptation

A No real solution
B x = −7, −2
C x = −2, 7
D x = −3.5

19.2 MCQ

Which option gives the complete real solution set?

$$x^2 + 2x - 24 = 0$$

Reveal · p. 713 · 2 · MCQ adaptation

A $x = 6, -4$
B $x = -4, 6$
C $x = -12, 2$
D $x = -6, 4$

19.7 MCQ

A quadratic graph touches the x-axis at exactly one point. How many distinct real zeros does it have?

Reveal · p. 713 · Application of 1 · Concept application

A 0
B 2
C 3
D 1

19.8 MCQ

Which values solve the equation from its x-intercepts?

$$y = (x - 2)(x + 3)$$

Reveal · p. 713 · Application of 1 · Concept application

A $\{-3, 2\}$
B $\{-2, 3\}$
C $\{2, 3\}$
D $\{-3, -2\}$

Q20 Complete the square

إكمال المربع

Remember First make the coefficient of x² equal to 1. Add (b/2)² to form a perfect square.

20.1 MCQ

Which option gives c so that the trinomial is a perfect square?

$$x^2 + 26x + c$$

Reveal · p. 729 · 1 · MCQ adaptation

A 26
B 676
C $\frac{169}{4}$
D 169

20.2 MCQ

Which option gives c so that the trinomial is a perfect square?

$$x^2 - 24x + c$$

Reveal · p. 729 · 2 · MCQ adaptation

A 576
B 36
C 144
D 24

20.3 MCQ

Which option gives c so that the trinomial is a perfect square?

$$x^2 - 19x + c$$

Reveal · p. 729 · 3 · MCQ adaptation

A $\frac{361}{16}$
B $\frac{361}{4}$
C 19
D 361

20.4 MCQ

Which option gives c so that the trinomial is a perfect square?

$$x^2 + 11x + c$$

Reveal · p. 729 · 4 · MCQ adaptation

A $\frac{121}{4}$
B 11
C 121
D $\frac{121}{16}$

02 / WRITE THE STEPS

Q21 · Written practice 1

Q21

Add and subtract polynomials

جمع كثيرات الحدود وطرحها

Remember Distribute a subtraction sign to every term; then collect like terms.

21.1 FRQ · show your working

Find the sum or difference.

$$(3c^3 - c + 11) - (c^2 + 2c + 8)$$

Reveal · p. 636 · 27

(a) Show your simplification.

21.2 FRQ · show your working

Find the sum or difference.

$$(z^2 + z) + (z^2 - 11)$$

Reveal · p. 636 · 28

(a) Show your simplification.

02 / WRITE THE STEPS

Q21 · Written practice 2

Q21

Add and subtract polynomials

جمع كثيرات الحدود وطرحها

Remember Distribute a subtraction sign to every term; then collect like terms.

21.3 FRQ · show your working

Find the sum or difference.

$$(2x - 2y + 1) - (3y + 4x)$$

Reveal · p. 636 · 29

(a) Show your simplification.

21.4 FRQ · show your working

Find the sum or difference.

$$(4a - 5b^2 + 3) + (6 - 2a + 3b^2)$$

Reveal · p. 636 · 30

(a) Show your simplification.

02 / WRITE THE STEPS

Q22 · Written practice 1

Q22

Multiply polynomials

ضرب كثيرات الحدود

Remember Multiply each term in the first polynomial by every term in the second; collect equal powers.

22.1 FRQ · show your working

Find the product.

$$(2y - 11)(y^2 - 3y + 2)$$

Reveal · p. 650 · 18

(a) Show all partial products and combine like terms.

22.2 FRQ · show your working

Find the product.

$$(4a + 7)(9a^2 + 2a - 7)$$

Reveal · p. 650 · 19

(a) Show all partial products and combine like terms.

02 / WRITE THE STEPS

Q22 · Written practice 2

Q22

Multiply polynomials

ضرب كثيرات الحدود

Remember Multiply each term in the first polynomial by every term in the second; collect equal powers.

22.3 FRQ · show your working

Find the product.

$$(m^2 - 5m + 4)(m^2 + 7m - 3)$$

Reveal · p. 650 · 20

(a) Show all partial products and combine like terms.

22.4 FRQ · show your working

Find the product.

$$(x^2 + 5x - 1)(5x^2 - 6x + 1)$$

Reveal · p. 650 · 21

(a) Show all partial products and combine like terms.

02 / WRITE THE STEPS

Q23 · Written practice 1

Q23

Multiplication properties of exponents

خواص ضرب القوى

Remember Add exponents in a product; multiply exponents in a power. Apply the power to the coefficient too.

23.1 FRQ · show your working

A square field has side length $5x^3y^5$. What is its area in terms of x and y?

Reveal · p. 576 · 25

(a) Show the area calculation.

23.2 FRQ · show your working

A cube-shaped shipping box has side length $4m^8n^4$. What is its volume in terms of m and n?



Reveal · p. 576 · 27

(a) Show the volume calculation.

02 / WRITE THE STEPS

Q23 · Written practice 2

Q23

Multiplication properties of exponents

خواص ضرب القوى

Remember Add exponents in a product; multiply exponents in a power. Apply the power to the coefficient too.

23.3 FRQ · show your working

Consider the two expressions.

$$(m^p)(m^q) \quad \text{and} \quad (m^p)^q$$

Reveal · p. 578 · 56

(a) What must be true about p and q for the expressions to be equal? Give an example.

(b) Assuming m, p and q are positive integers, can $(m^p)(m^q) > (m^p)^q$ ever be true? Explain.

23.4 FRQ · show your working

For nonzero real a and b and integers m and t, simplify and explain.

$$\left(-\frac{a^m}{b^t}\right)^{2t}$$

Reveal · p. 578 · 57

(a) Apply the power rules and explain the sign.

02 / WRITE THE STEPS

Q23 · Written practice 3

Q23

Multiplication properties of exponents

خواص ضرب القوى

Remember Add exponents in a product; multiply exponents in a power. Apply the power to the coefficient too.

23.5 FRQ · show your working

Consider y=x, y=x² and y=x³.

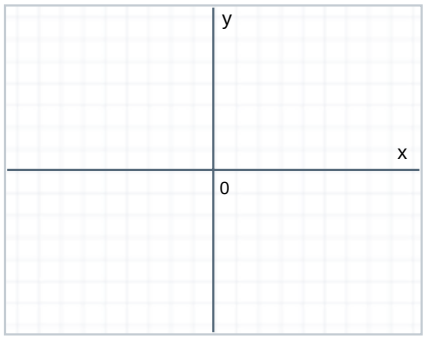
Reveal · p. 578 · 58

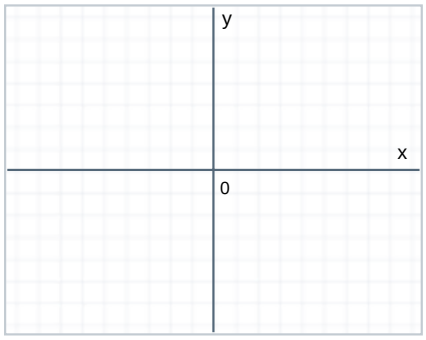
fx-991EX TABLE: MENU → 9; choose f(x) only in SETUP. For y=X, use X=−1,0,1 (3 points). For y=X², use X=−2,−1,0,1,2 (5 points). Set Start, End, Step; copy pairs and draw manually. Repeat for X³.

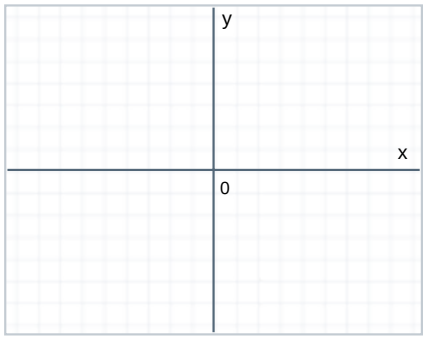
Equation	Related expression	Power of x	Linear / nonlinear
$y = x$			
$y = x^2$			
$y = x^3$			

(a) For each equation, write the related expression and record the power of x.

(b) Graph each equation using a graphing calculator.







x			
y			

x					
y					

x					
y					

(c) Classify each graph as linear or nonlinear.

(d) Explain how to determine whether an equation, or its related expression, is linear or nonlinear without graphing.

02 / WRITE THE STEPS

Q24 · Written practice 1

Q24

Quadratic extrema and graph features

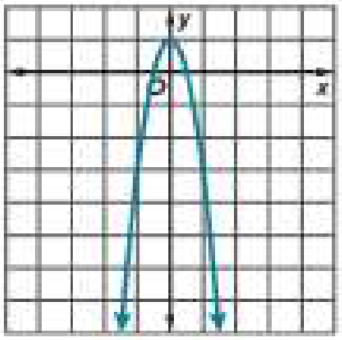
القيم العظمى والصغرى وخصائص المنحنى

Remember The vertex (h,k) gives the minimum k when the graph opens up and maximum k when it opens down; the axis is x=h.

24.1 FRQ · show your working

Identify the axis of symmetry, vertex and y-intercept of the graph. Then describe the end behavior.

Reveal · p. 695 · 1



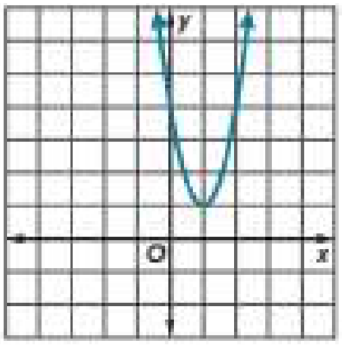
(a) Axis of symmetry and vertex.

(b) y-intercept and end behavior.

24.2 FRQ · show your working

Identify the axis of symmetry, vertex and y-intercept of the graph. Then describe the end behavior.

Reveal · p. 695 · 2



(a) Axis of symmetry and vertex.

(b) y-intercept and end behavior.

02 / WRITE THE STEPS

Q24 · Written practice 2

Q24

Quadratic extrema and graph features

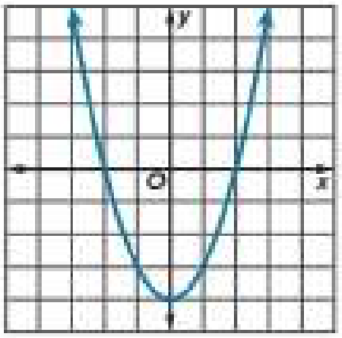
القيم العظمى والصغرى وخصائص المنحنى

Remember The vertex (h,k) gives the minimum k when the graph opens up and maximum k when it opens down; the axis is x=h.

24.3 FRQ · show your working

Identify the axis of symmetry, vertex and y-intercept of the graph. Then describe the end behavior.

Reveal · p. 695 · 3



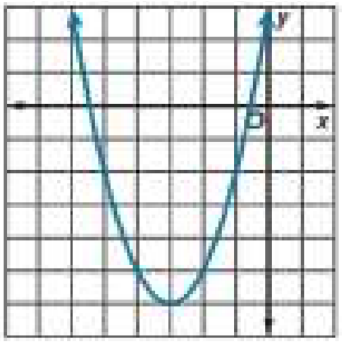
(a) Axis of symmetry and vertex.

(b) y-intercept and end behavior.

24.4 FRQ · show your working

Identify the axis of symmetry, vertex and y-intercept of the graph. Then describe the end behavior.

Reveal · p. 695 · 4



(a) Axis of symmetry and vertex.

(b) y-intercept and end behavior.

02 / WRITE THE STEPS

Q25 · Written practice 1

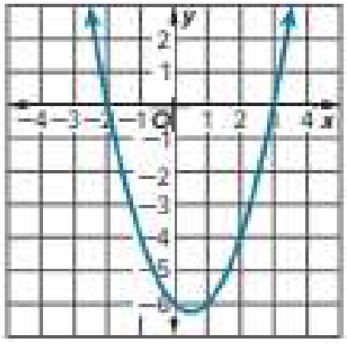
Q25 **Write a quadratic from its graph**
كتابة الدالة التربيعية من الرسم

> **Remember** Read the zeros r,s, write $y=a(x-r)(x-s)$, then use another point to find a. Expand to $y=ax^2+bx+c$.

25.1 FRQ · show your working

Write a quadratic function for the given graph. Give the equation in standard form.

Reveal · p. 722 · 31



(a) Show how you find the equation.

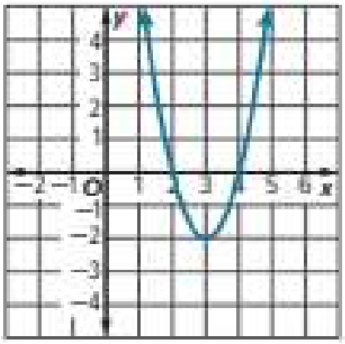
fx-991EX · TABLE MENU → Table (9). SETUP → Table → f(x) only. Enter your equation using ALPHA X, then =. Set Start, End, Step; copy 5 pairs. Include the vertex and two points on each side. Find the equation from the original graph before using TABLE.

x					
y					

25.2 FRQ · show your working

Write a quadratic function for the given graph. Give the equation in standard form.

Reveal · p. 722 · 32



(a) Show how you find the equation.

fx-991EX · TABLE MENU → Table (9). SETUP → Table → f(x) only. Enter your equation using ALPHA X, then =. Set Start, End, Step; copy 5 pairs. Include the vertex and two points on each side. Find the equation from the original graph before using TABLE.

x					
y					

02 / WRITE THE STEPS

Q25 · Written practice 2

Q25 Write a quadratic from its graph

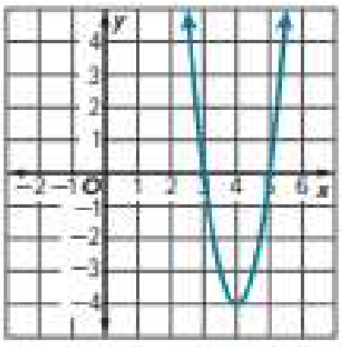
كتابة الدالة التربيعية من الرسم

Remember Read the zeros r,s, write $y=a(x-r)(x-s)$, then use another point to find a. Expand to $y=ax^2+bx+c$.

25.3 FRQ · show your working

Write a quadratic function for the given graph. Give the equation in standard form.

Reveal · p. 722 · 33



(a) Show how you find the equation.

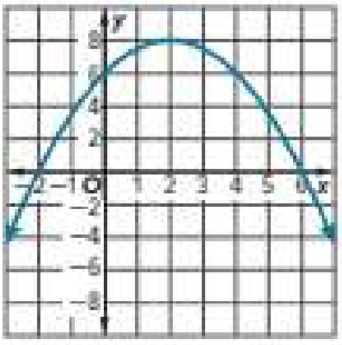
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x					
y					

25.4 FRQ · show your working

Write a quadratic function for the given graph. Give the equation in standard form.

Reveal · p. 722 · 34



(a) Show how you find the equation.

fx-991EX · TABLE MENU → Table (9). SETUP → Table → f(x) only. Enter your equation using ALPHA X, then =. Set Start, End, Step; copy 5 pairs. Include the vertex and two points on each side. Find the equation from the original graph before using TABLE.

x					
y					

03 / CHECK & LEARN

MCQ answer key

Check after solving. Match the printed question number to its correct option. Full explanations are available in the online LO bank.

Q1 · FOIL binomial products

1.1: C 1.2: B 1.3: A 1.4: D

Q2 · Squares of sums and differences

2.1: B 2.2: A 2.7: A 2.8: C

Q3 · Sum and difference products

3.1: A 3.2: D 3.3: C 3.4: B

Q4 · Factor using distribution

4.1: D 4.2: C 4.3: B 4.4: A

Q5 · Analyse monic trinomial factors

5.2: B 5.3: A 5.5: C 5.6: B

Q6 · Factor non-monic trinomials

6.1: B 6.2: A 6.3: D 6.4: C

Q7 · Difference of squares

7.1: A 7.2: D 7.3: C 7.4: B

Q8 · Divide monomials

8.2: C 8.3: B 8.7: B 8.8: D

Q9 · Negative and zero exponents

9.1: C 9.2: B 9.7: A 9.8: C

Q10 · Rational exponents

10.1: B 10.2: A 10.5: B 10.6: A

Q11 · Products of square roots

11.1: A 11.2: D 11.3: C 11.4: B

Q12 · Quotients of radicals

12.1: D 12.2: C 12.3: B 12.4: A

Q13 · Add and subtract radicals

13.1: C 13.2: B 13.5: C 13.6: B

Q14 · Factoring in context

14.1: B 14.2: A 14.3: D 14.6: A

Q15 · Square root and completing-square methods

15.1: A 15.2: D 15.7: B 15.8: B

Q16 · Translations of quadratics

16.1: D 16.2: C 16.7: A 16.8: D

Q17 · Factoring quadratic equations

17.1: C 17.2: B 17.3: A 17.4: D

Q18 · Dilations and reflections of quadratics

18.1: B 18.2: A 18.7: B 18.8: A

Q19 · Solve quadratics by graphing

19.1: A 19.2: D 19.7: D 19.8: A

Q20 · Complete the square

20.1: D 20.2: C 20.3: B 20.4: A

03 / CHECK & LEARN

Worked solutions 1

21.1 p. 636 · 27

(a) Correct sign distribution and like-term collection (6 marks); simplified polynomial 3·c³ - c² - 3·c + 3 (4 marks).

$$3c^3 - c^2 - 3c + 3$$

21.2 p. 636 · 28

(a) Correct sign distribution and like-term collection (6 marks); simplified polynomial 2·z² + z - 11 (4 marks).

$$2z^2 + z - 11$$

21.3 p. 636 · 29

(a) Correct sign distribution and like-term collection (6 marks); simplified polynomial -2·x - 5·y + 1 (4 marks).

$$-2x - 5y + 1$$

21.4 p. 636 · 30

(a) Correct sign distribution and like-term collection (6 marks); simplified polynomial 2·a - 2·b² + 9 (4 marks).

$$2a - 2b^2 + 9$$

22.1 p. 650 · 18

(a) All partial products (6 marks); correct collection of like terms (4 marks): 2·y³ - 17·y² + 37·y - 22.

$$2y^3 - 17y^2 + 37y - 22$$

22.2 p. 650 · 19

(a) All partial products (6 marks); correct collection of like terms (4 marks): 36·a³ + 71·a² - 14·a - 49.

$$36a^3 + 71a^2 - 14a - 49$$

22.3 p. 650 · 20

(a) All partial products (6 marks); correct collection of like terms (4 marks): m⁴ + 2·m³ - 34·m² + 43·m - 12.

$$m^4 + 2m^3 - 34m^2 + 43m - 12$$

03 / CHECK & LEARN

Worked solutions 2

22.4 p. 650 · 21

(a) All partial products (6 marks); correct collection of like terms (4 marks): 5·x⁴ + 19·x³ - 34·x² + 11·x - 1.

$$5x^4 + 19x^3 - 34x^2 + 11x - 1$$

23.1 p. 576 · 25

(a) A=s²=(5x³y⁵)²=25x⁶y¹⁰. Square the coefficient (3), multiply each exponent by 2 (5), final expression (2).

$$A = 25x^6y^{10}$$

23.2 p. 576 · 27

(a) V=s³=(4m⁸n⁴)³=64m²⁴n¹². Cube the coefficient (3), multiply each exponent by 3 (5), final expression (2).

$$V = 64m^{24}n^{12}$$

23.3 p. 578 · 56

(a) For positive m≠1, require p+q=pq, or (p−1)(q−1)=1 (3). Example p=q=2 (2). Without that assumption, m=1 makes both expressions equal for all defined p,q; other bases require domain care.

$$p + q = pq \quad (m > 0,\ m \neq 1)$$

(b) Yes. Take m=2 and p=q=1: 2·2=4>2¹=2 (3). For m>1 compare exponents p+q and pq (2).

$$2^1 \cdot 2^1 = 4 > 2^{1\cdot 1} = 2$$

23.4 p. 578 · 57

(a) Since 2t is even, (−1)²ᵗ=1 (3). Multiply the numerator exponents to get 2mt (3), and denominator exponents to get 2t² (3); final quotient (1). Nonzero a,b make all integer powers defined.

$$\frac{a^{2mt}}{b^{2t^2}}$$

23.5 p. 578 · 58

(a) x=x¹ (power 1), x·x=x² (power 2), x·x·x=x³ (power 3). One mark per row.

(b) Correct line, upward parabola and increasing cubic through the origin. One mark per graph.

(c) y=x is linear; y=x² and y=x³ are nonlinear. One mark for all three correct.

(d) After simplification, y=ax+b is linear (2); higher powers of x such as x² and x³ are nonlinear (1).

TABLE: (-2, 4) · (-1, 1) · (0, 0) · (1, 1) · (2, 4)

24.1 p. 695 · 1

(a) Axis x=0 (2); vertex (0,1) (3).

(b) y-intercept (0,1) (2); y→−∞ as x→±∞; maximum 1 (3). The extremum is the vertex y-coordinate.

03 / CHECK & LEARN

Worked solutions 3

24.2 p. 695 · 2

(a) Axis x=1 (2); vertex (1,1) (3).

(b) y-intercept (0,4) (2); y → +∞ as x → ±∞; minimum 1 (3). The extremum is the vertex y-coordinate.

24.3 p. 695 · 3

(a) Axis x=0 (2); vertex (0,−4) (3).

(b) y-intercept (0,−4) (2); y → +∞ as x → ±∞; minimum −4 (3). The extremum is the vertex y-coordinate.

24.4 p. 695 · 4

(a) Axis x=−3 (2); vertex (−3,−6) (3).

(b) y-intercept (0,3) (2); y → +∞ as x → ±∞; minimum −6 (3). The extremum is the vertex y-coordinate.

25.1 p. 722 · 31

(a) Zeros −2 and 3 give y=a(x+2)(x−3). The y-intercept −6 gives a=1. Correct reading of two graph features (4), coefficient a (3), expanded standard form (3).

$$y = x^2 - x - 6$$

TABLE: (-1.5, -2.25) · (-0.5, -5.25) · (0.5, -6.25) · (1.5, -5.25) · (2.5, -2.25)

25.2 p. 722 · 32

(a) Zeros 2 and 4 give y=a(x−2)(x−4). The vertex (3,−2) gives a=2. Correct reading of two graph features (4), coefficient a (3), expanded standard form (3).

$$y = 2x^2 - 12x + 16$$

TABLE: (1, 6) · (2, 0) · (3, -2) · (4, 0) · (5, 6)

25.3 p. 722 · 33

(a) Zeros 3 and 5 give y=a(x−3)(x−5). The vertex (4,−4) gives a=4. Correct reading of two graph features (4), coefficient a (3), expanded standard form (3).

$$y = 4x^2 - 32x + 60$$

TABLE: (2, 12) · (3, 0) · (4, -4) · (5, 0) · (6, 12)

25.4 p. 722 · 34

(a) Zeros −2 and 6 give y=a(x+2)(x−6). The vertex (2,8) gives a=−½. Correct reading of two graph features (4), coefficient a (3), expanded standard form (3).

$$y = -\tfrac{1}{2}x^2 + 2x + 6$$

TABLE: (0, 6) · (1, 7.5) · (2, 8) · (3, 7.5) · (4, 6)